

Programming Project / C++ – Bank Management System

Student 1: Florian Dumitrescu

Student 2: Ioan Mitrut

I. Task Description

Student 1 – responsible for:

- Entity, Client, Account, CurrentAccount, SavingsAccount modeling
- deposit, withdraw, print services.
- writing objects to files
- cmdline parsing commands: --deposit, --withdraw, --list-owner-acc, --show-account

Student 2 - responsible for:

- Bank, BankController operations.
- reading objects from files
- cmdline parsing commands: --add-client, --open-account, --close-account

II. Data structures

```
Bank{
    -clients: vector<shared_ptr<Client>>
    -accounts: vector<shared_ptr<Account>>
}
Entity{
    - id: int
    - nextId: static int
}
Account: public Entity{
    - id: int
    - owner: shared_ptr<Client>
    - iban: string
    - balance: double
    - currency: string
}
CurrentAccount: public Account{
    - managementFee: double
}
SavingsAccount: public Account{
    - interestRate: double
}
Client: public Entity{
    - id: int
    - name: string
}
```

III. File Structure

accounts.csv
id, ownerId, type, iban, currency, balance, accountSpecificValue

clients.csv
id, name

IV. Interacting with executable

Application will offer the interaction between its components as follows:

`./bankapp.exe --add-client <clientName>`

In order for the bank to add a new client.

`./bankapp.exe --open-account <clientName> <accountType> <currency>`

In order for the bank to open an account of certain type and currency for the provided client by name.

`./bankapp.exe --close-account <clientName> <IBAN>`

In order for the bank to close a given account for a given client, provided client and account exist and client is the owner of the account.

`./bankapp.exe --list-owner-acc <clientName>`

To list the accounts that belong to a client.

`./bankapp.exe --show-account <IBAN>`

To query a single account's details.

`./bankapp.exe client --deposit <clientName> <IBAN> <amount>`

In order for a client to deposit money in an owned account.

`./bankapp.exe client --withdraw <clientName> <IBAN> <amount>`

In order for a client to withdraw money from an owned account.